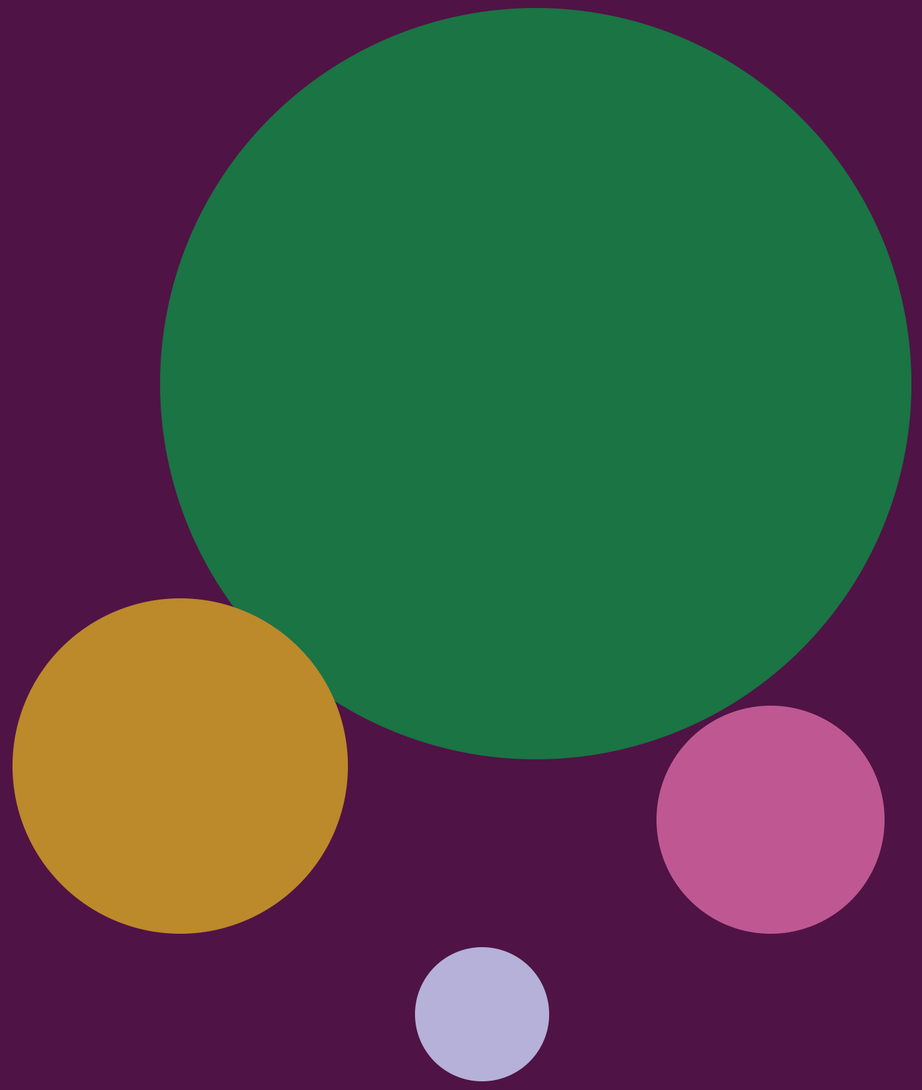
TPU-DISTIL · OCTOBER 2026

Big-model quality at a small-model price

How we taught small models to do the job of a 552B model on SQL, finance and maths, and served each one on a single Cloud TPU chip

Jawad Amin and Fred

github.com/jawadaminGOOG/tpu-distil



AGENDA

What we'll cover

01

The problem

Why big models are hard to serve and small models fall short

~6 MIN

02

The experiments

The three tasks, the two students and how we chose them

~5 MIN

03

The journey

Our software stack and the six training steps, with what we learned at each one

~11 MIN

04

What we got

The results, the two recipes and why we trust the numbers

~11 MIN

05

Fred's experiment

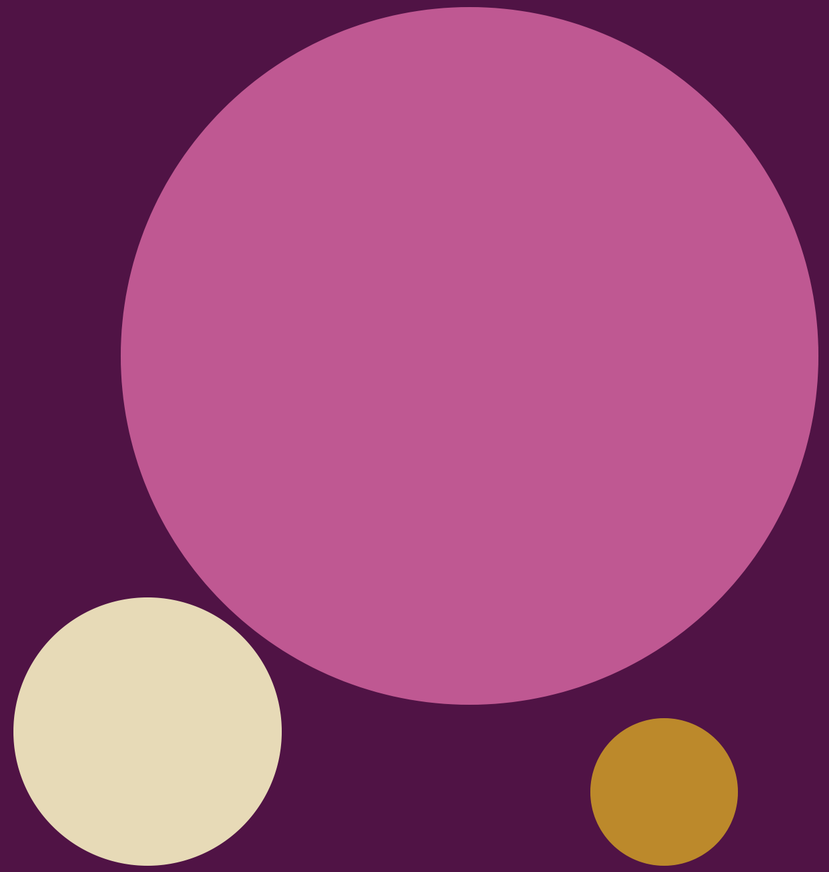
How Fred taught models to draw

~7 MIN

01

The problem

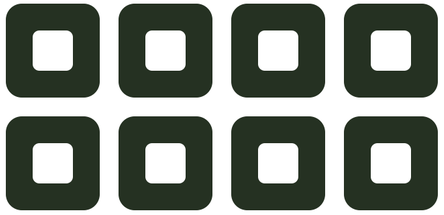
Frontier models do the job well, but they are expensive and GPUs are hard to get. Small models are cheap, but they don't know the domain.



THE PROBLEM

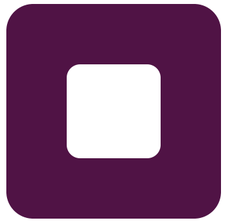
Big models are expensive; small ones fall short

One frontier model needs a whole GPU server



8× H200 for one teacher replica

VS



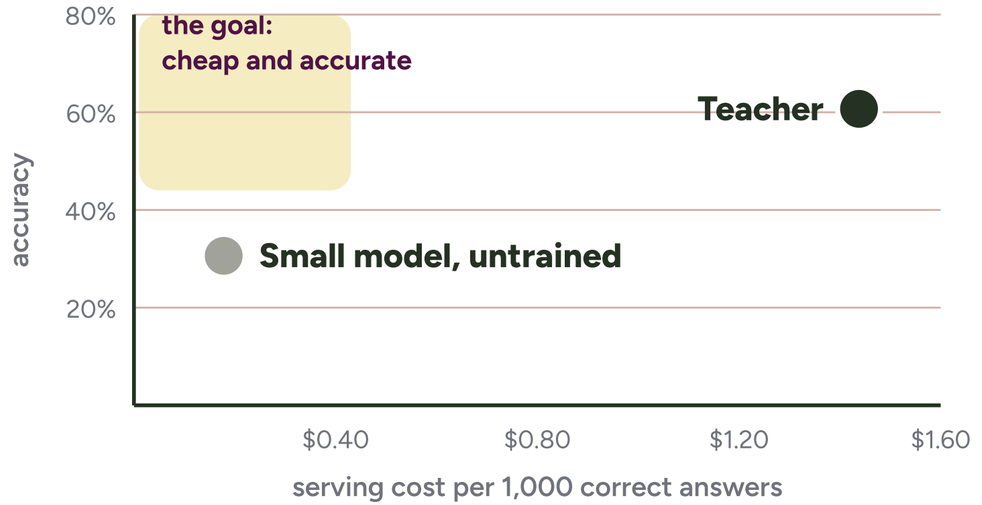
1× TPU v6e chip for the student

552B parameters in the teacher, DeepSeek-V4.1-Flash

$36.60 / h for one teacher replica, at the cheapest listed H200 rate

$2.70 / h for one TPU v6e chip, on demand

Every correct answer has a price



The big model is accurate but expensive

It writes correct SQL for **60.7%** of questions, and 1,000 correct answers cost **$1.44**.

The small model is cheap but often wrong

1,000 correct answers cost only **$0.18**, but it gets just **30.6%** of questions right.

BIRD dev: 1,534 questions on 11 databases never used in training. Cost = $ per hour ÷ (requests per second × 3,600 × accuracy).

THE PROBLEM

Generic small models miss domain details

QUESTION

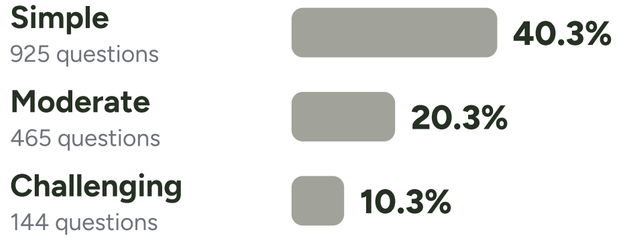
Which county is member Adela O'Gallagher's hometown in?

UNTRAINED · SQLITE SYNTAX ERROR

```
WHERE member.first_name = 'Adela'
  AND member.last_name = 'O\'Gallagher'
```

TRAINED STUDENT · CORRECT ROWS

```
WHERE m.first_name = 'Adela'
  AND m.last_name = 'O''Gallagher'
```



WHY FINE-TUNE FOR THE DOMAIN

The untrained 1.7B model gets only **10%** of the hardest questions right. Of its wrong answers, **574** return the wrong rows and **462** are not valid SQLite, because the model invents columns or uses another dialect's syntax. Companies see the same problem with their own schemas and SQL dialects.

THE BET

We bet that a big model could teach a small one, and that the small one could run on a single TPU chip.

Before we started, we set two targets: at least **5 points** better than the untrained student, and at least **10×** cheaper per correct answer than the teacher.

BIRD dev; accuracy by BIRD's difficulty label for the untrained Qwen3-1.7B; example dev-1347, question paraphrased

02

The experiments

We ran the same pipeline on three tasks with two different students.

THE EXPERIMENTS

Each task has an automatic grader

BIRD TEXT-TO-SQL

INPUT a database schema, example rows, a hint and a question

ANSWER one SQLite query

GRADER runs the query and compares its rows with the gold rows

EVALUATE dev: 1,534 questions on 11 unseen databases

TRAIN 8,432 questions on 64 databases

STUDENT **Qwen3-1.7B** scores 30.6%; the teacher scores 60.7%

FinQA FINANCIAL REPORTS

INPUT one earnings-report page: text, a table, a question

ANSWER `Answer: <number>`

GRADER checks that the number is within 1% of the gold answer

EVALUATE test: 1,147 questions on unseen report pages

TRAIN 6,109 questions after removing overlaps

STUDENT **Qwen3-1.7B** scores 44.0%; the teacher scores 75.7%

MATH COMPETITION MATHS

INPUT a competition problem, 7 subjects, 5 levels

ANSWER `\boxed{answer}`

GRADER checks the boxed answer against the gold answer after standard clean-up

EVALUATE test: 5,000 problems

TRAIN 6,262 problems after removing overlaps

STUDENT **Gemma 4 E2B** scores 74.9%; the teacher scores 92.5%

One student starts weak, the other starts strong

untrained student vs teacher, same prompt

Qwen3-1.7B starts weak

It is about 30 points behind the teacher on SQL and finance, so it has the most to learn. It is also the cheapest to serve.

Gemma 4 E2B starts strong

It already solves 74.9% of MATH problems. We wanted to know if a teacher still helps.

Teacher DeepSeek-V4.1-Flash on 8× H200 for all three; students served by vLLM-TPU on one v6e chip; non-thinking mode throughout
Untrained student vs teacher, same prompt, greedy decoding; Qwen3-4B already scored 61.7% on BIRD and was not used

THE EXPERIMENTS

Six decisions shaped the experiments

TASKS

Tasks with automatic graders

A program checks every answer, so one run takes minutes and can show a 2-point change. An agentic benchmark needs hours per run.

STUDENTS

A weak student and a strong one

With two students we can see when a teacher helps, and when it doesn't.

TRAINING DATA

Only verified answers

The grader checks every training answer, because a wrong answer would teach the student the wrong thing.

FORMAT

Short answers

Teacher and student both reason briefly. Without long reasoning traces, the teacher samples 4× faster.

TRAINING

LoRA on one TPU host

LoRA fits on a single TPU v6e-8, and every size is a multiple of 256 to match the chip's matrix unit.

EVALUATION

Comparisons within one run

We evaluate every model, a control and a second copy of the baseline together.

03

The journey

The software we used, the six training steps, and what we learned at each step.

THE JOURNEY

Every experiment follows the same six steps

1 Measure
graders, noise floor
teacher vs student

→

2 Generate
teacher answers, or
the student's own

→

3 Verify
grade every answer
audit for leaks

→

4 Distil
LoRA SFT on TPU
10–27 minutes

→

5 Reinforce
GRPO, 6 rounds
rewards correct answers

→

6 Serve
1 chip per replica
price each answer

We built everything on open-source libraries

distilling takes 10–27 min on one TPU v6e-8 host

Teacher	8× H200, Vertex AI	DeepSeek-V4.1-Flash · Model Garden vLLM · TP 8 + expert parallel
Pipeline	CPU VM	Python standard library · sqlite3 · JSONL records · paired bootstrap
Student serving	1 TPU v6e chip each	vLLM-TPU · tpu-inference backend · JAX
Training	TPU v6e-8 host	JAX + libtpu · Tunix · qwix LoRA · Flax NNX · Optax · Orbax
Glue	everywhere	Hugging Face transformers · safetensors · Cloud Storage

One code path Both students share one code path; only the task module changes.

One checkpoint format Trainer, server and storage all use merged Hugging Face safetensors.

Simple pipeline scripts It uses standard-library Python and JSONL, so every number can be regenerated.

One trainer for both models The same Tunix trainer handles Qwen3 and Gemma 4 on one v6e-8.

Versions: vLLM 0.26.1rc1 TPU build, JAX 0.11.0, libtpu 0.0.44, Tunix 0.1.8, qwix 0.1.8, Flax 0.12.8, transformers 5.12.1 (docs/RECIPE.md)

THE JOURNEY · STEPS 1–2 OF 6

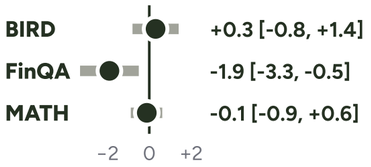
Measure the noise, then choose the training answers

1 MEASURE Check the measurements before the experiment

Is the grader right?

	gold answers	shuffled answers
BIRD	100%*	0.2%
FinQA	100%	0.6%
MATH	100%	0.0%

How much does serving noise move the score?



Teacher − untrained:
BIRD +30.1 · FinQA +31.7 · MATH +17.6 points

WHAT WE DID We checked that every grader gives gold answers 100% and shuffled answers close to 0%. We also served two copies of each untrained model to measure noise.

DECISION We compare models only **within a single run**, and every run includes a second copy of the baseline.

WHAT WE LEARNED Greedy decoding on vLLM-TPU isn't perfectly repeatable. Two copies of the same FinQA model differed by 1.9 points, so we can't trust a small gain without a noise floor.

2 GENERATE Choose who writes the training answers

Teacher
8× H200

× 4 →

Teacher answers, T 0.7
BIRD 33,728 · FinQA 24,436 · MATH 25,048
26–38 minutes per task on 8 GPUs
for a weak student

Student
1 v6e chip each

× 8 →

Its own answers, T 0.7
MATH 50,096 · the control on every task
no teacher time
for a strong student

WHAT WE DID We sampled the teacher 4 times and the student 8 times per question, using training questions only. Both write the same short format: brief reasoning, then one answer.

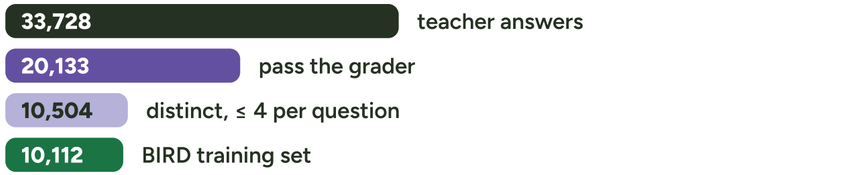
DECISION A weak student learns from the teacher's answers. A strong student learns from **its own verified answers**, which keep its style of reasoning.

WHAT WE LEARNED The training answers should look like what the student will write when it serves. On MATH, the teacher's shorter solutions made a strong student worse.

* BIRD gold scores 100% of the 1,530 gold queries that execute (99.7% of 1,534)

Verify every answer, then distil on one TPU host

3 VERIFY Keep only the answers the grader accepts



TASK	ANSWERS	PASS THE GRADER	TRAINING SET
FinQA, teacher	24,436	18,487	6,016
MATH, student	50,096	72.6%	4,992

WHAT WE DID We graded every answer and kept only the correct ones, with up to four distinct answers per question. A separate audit re-checked every row and looked for overlap with the evaluation sets.

DECISION We train **only on verified answers**, and we reject any row that comes from an evaluation database, report page or problem.

WHAT WE LEARNED The grader turned out to be the most valuable code we wrote, because 40% of the teacher's BIRD answers were wrong. The audit found **no leaks** on any task.

4 DISTIL Distil the student on one TPU host

LoRA fine-tuning on TPU v6e-8



117 / 69 / 57 steps (BIRD / FinQA / MATH) · 10–27 minutes

Checks we ran before trusting any result

0.998
trainer and server agree (top-1)

1 : 1
retrain gave a byte-identical model

rank 256 · alpha 512 · attention + MLP · batch 256 · 3 epochs
Qwen3 and Gemma 4 loaders · loss on answer tokens only

WHAT WE DID We trained low-rank adapters on all eight chips of one TPU v6e-8 host, merged them into the model and exported a standard checkpoint for serving. Every task used the same settings.

DECISION We used LoRA instead of full fine-tuning because it fits on one host. Every size is a multiple of 256 to match the TPU's matrix unit.

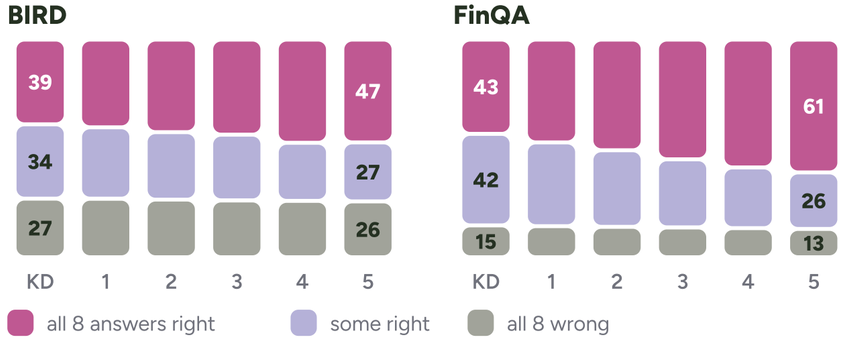
WHAT WE LEARNED Before you trust a result, check that the trainer and the server compute the same model. Keep adapter weights in float32, because in bf16 the optimizer's first update divides zero by zero.

Step 3 bars: BIRD; table: FinQA and MATH (docs/RECIPE.md)

THE JOURNEY · STEPS 5–6 OF 6

Reinforce with GRPO, then serve on a single chip

5 REINFORCE RL makes sometimes-right answers always right

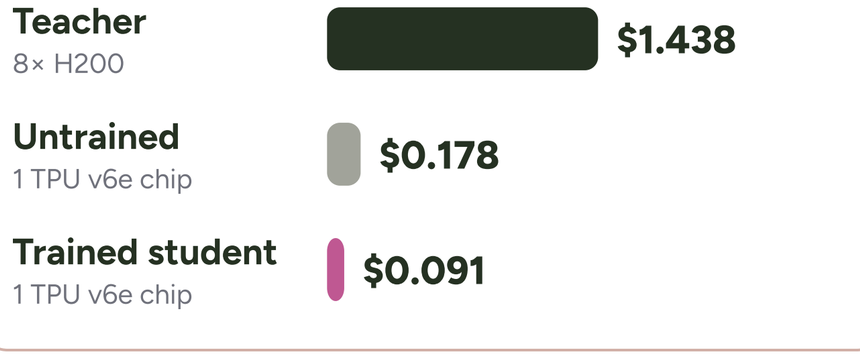


WHAT WE DID We ran six rounds of GRPO. The student answers each training question 8 times, gets a reward of 1 when the grader accepts an answer, and moves toward the better answers in each group.

DECISION The update learns only from questions with **mixed outcomes**, because all-right and all-wrong groups give it nothing to compare. A small KL penalty keeps the student close to where it started.

WHAT WE LEARNED On FinQA, the share of questions the student always gets right grew from 43% to 61%, while the share it never gets right barely moved. On held-out data, RL added 4.0 points on BIRD, 5.8 on FinQA and 1.8 on MATH.

6 SERVE Each student runs on a single chip



WHAT WE DID We served every model with the same flags, one vLLM-TPU copy per chip, and raised the load until errors appeared. Then we priced the best error-free rate.

DECISION We measure the **cost of each correct answer**, because both accuracy and throughput affect it. Tokens per second would ignore accuracy.

WHAT WE LEARNED Answer length drives throughput. The trained BIRD student is faster than the untrained one, at 15.8 requests per second, but the MATH student's long answers make it slower than the teacher.

Step 5: training pool, 8 samples per question at T 1.0, policy that produced each round's samples; BIRD all-wrong = 100 − pass@8, all-right = pass@8 − mixed
Step 6: serving cost per 1,000 correct BIRD answers; student at the on-demand TPU v6e rate, teacher at the cheapest listed H200 rate

04

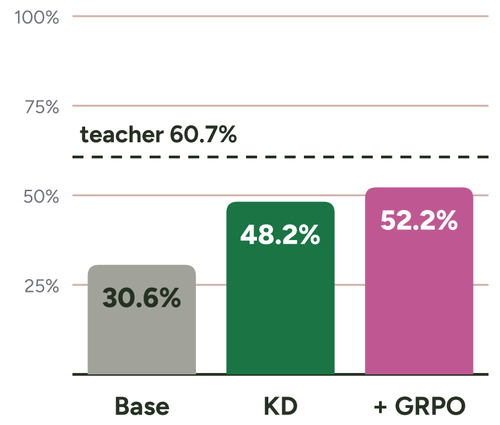
What we got

What each experiment achieved, why the recipe changed for the strong student, and why we trust the numbers.

WHAT WE GOT

All three students improved

BIRD SQL · +21.6 points

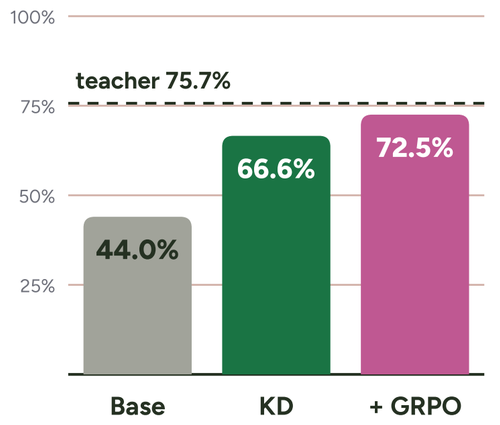


+17.6 points from distillation, 95% CI [+15.2, +20.1]

+4.0 more points from GRPO, 95% CI [+2.2, +5.9]

72% of the gap to the teacher closed, for $136 of compute

FinQA finance · +28.4 points

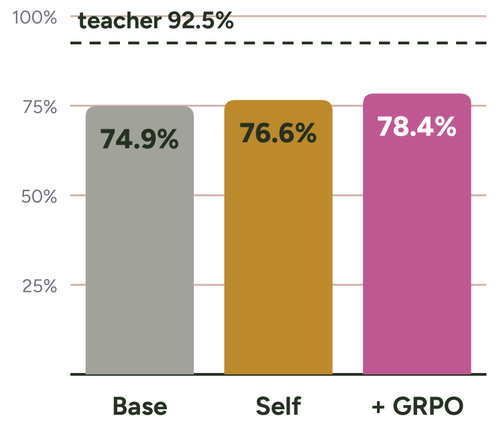


+22.6 points from distillation, 95% CI [+19.4, +25.7]

+5.8 more points from GRPO, 95% CI [+3.9, +7.8]

90% of the gap to the teacher closed, for $79 of compute

MATH Gemma · +3.5 points



+3.5 points from self-distillation and GRPO, 95% CI [+2.6, +4.4]

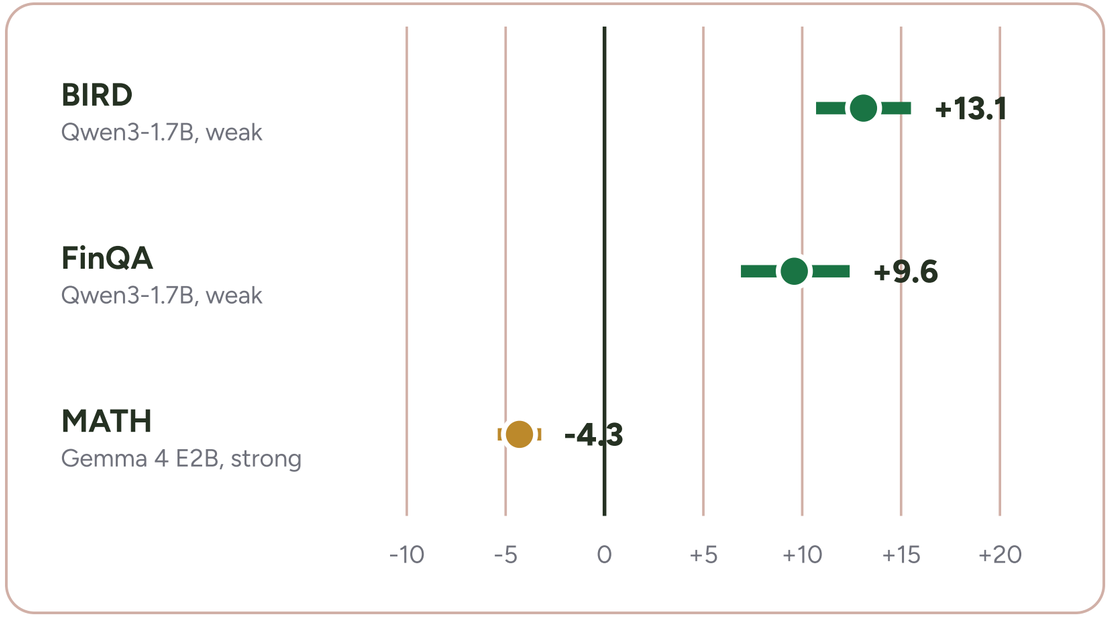
+1.82 more points from GRPO, 95% CI [+1.00, +2.66]

20% of the gap to the teacher closed, for $159 of compute

Each panel: one same-run evaluation; KD = teacher answers, Self = the student's own verified answers; dashed line = teacher; paired differences with bootstrap 95% CIs
BIRD dev, 1,534 questions; FinQA test, 1,147 questions, settings and schedule unchanged from BIRD; MATH test, 5,000 problems

WHAT WE GOT

The control experiment tells us which recipe to use

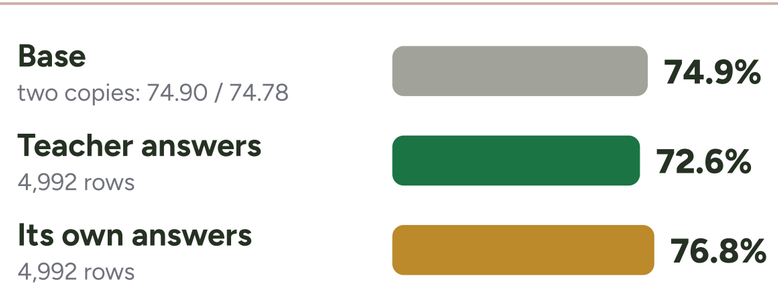


What the control is. We train the same SFT recipe on the same number of rows, using the student's own verified answers instead of the teacher's.

The weak student learns more from the teacher. The teacher's answers beat the student's own by 13.1 points on BIRD and 9.6 on FinQA.

The strong student learns more from itself. On MATH, the teacher's answers score 4.3 points lower than the student's own.

A strong student lost accuracy copying the teacher



597 vs 1,027 average tokens per answer. The distilled student copied the teacher's shorter solutions.

−3.3 / −4.1 points lost on the two hardest levels, where short solutions hurt most

−10.8 points when the teacher corrected the student's mistakes instead

LESSON

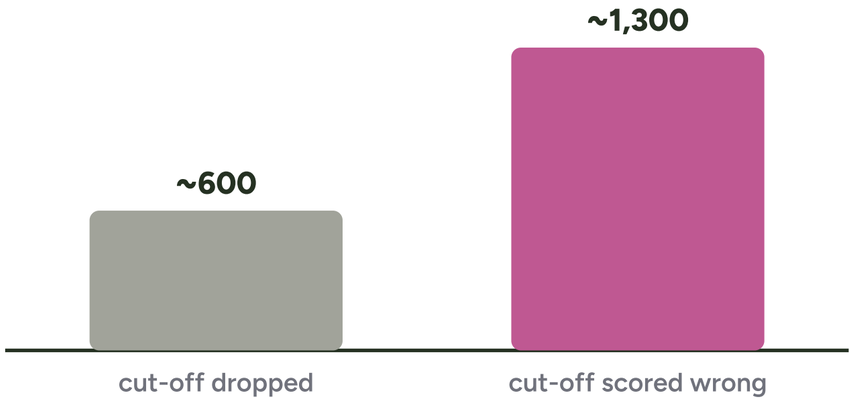
Gemma already reasons well in its own style. Copying the teacher's shorter style cost it accuracy, while training on its own answers improved it.

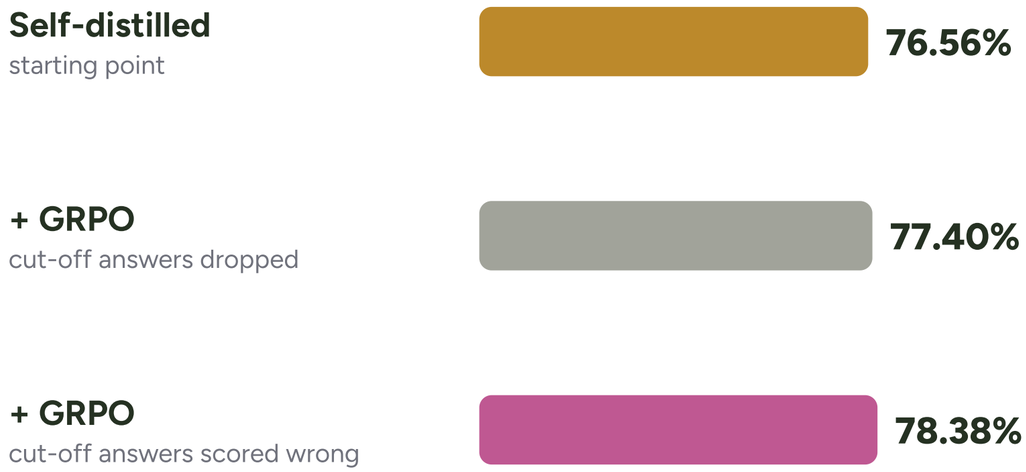
Left: teacher-answer SFT minus self-distillation SFT, paired differences in points with bootstrap 95% CIs, same-run evaluations
Right: MATH test, 5,000 problems, same-run evaluation; teacher corrections from a separate same-run evaluation

WHAT WE GOT

Counting cut-off answers as wrong made RL work

Problems per round with mixed outcomes





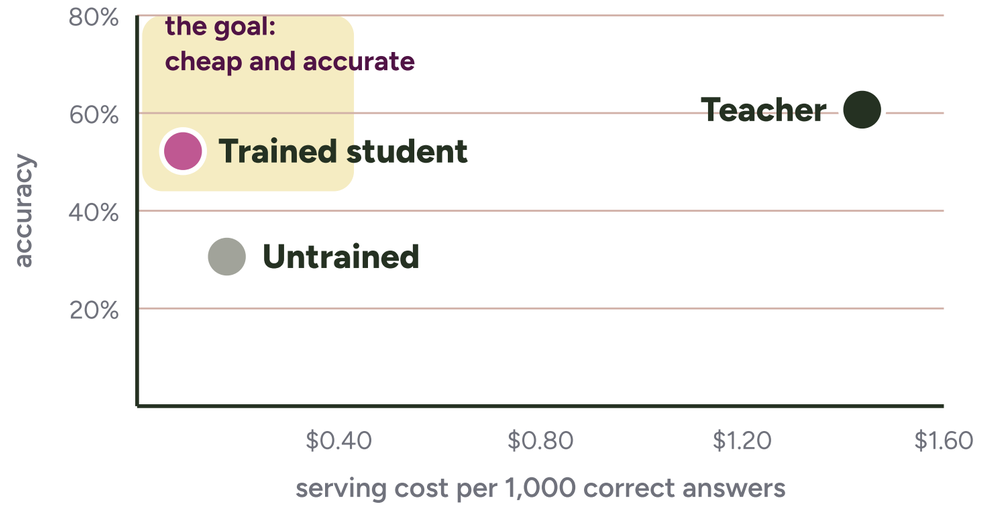
LESSON

About 15% of the student's MATH answers hit the 2,048-token limit before giving an answer, and our original GRPO threw them away. Once we scored them as wrong, the update had twice as many problems to learn from, and GRPO added **1.82** points [+1.00, +2.66] instead of 0.84.

MATH test, 5,000 problems, same-run evaluation; 6 GRPO rounds each; cut-off answers keep reward 0 and no end-of-turn token in the loss

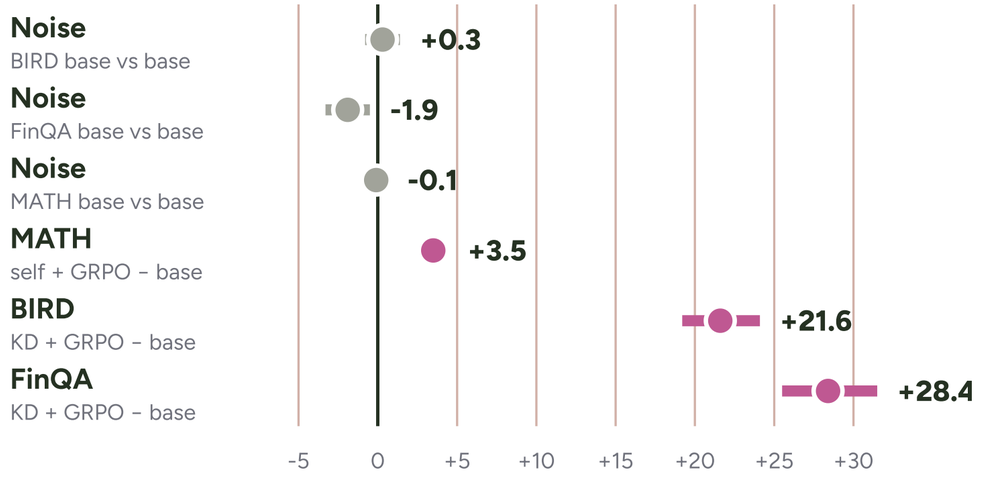
WHAT WE GOT

The trained student is both cheap and accurate



TASK	PRICE	× SPEED	× ACCURACY	= CHEAPER
BIRD	13.6×	1.35×	0.86×	**15.8×**
FinQA	13.6×	1.35×	0.96×	**17.5×**
MATH	13.6×	0.42×	0.85×	**4.8×**

Why we trust these results



Repeat runs agree. Three BIRD evaluations gave +17.8, +18.1 and +17.6. Four MATH evaluations stayed between +1.3 and +1.9.

No evaluation data leaked. A row-by-row audit found no evaluation database, report page or problem in the training data.

The results reproduce. A retrain produced a byte-identical model, and one script regenerates every number.

Chart: BIRD. Teacher at its cheapest listed H200 rate, students at the on-demand TPU rate; cost ratio = 13.6 × throughput × accuracy
Right: paired differences in points with bootstrap 95% CIs; each row from a same-run evaluation

05

Fred teaches models to draw

Fred took a different route to the same goal: a small model that does a specialist job well.

FRED'S EXPERIMENT

The problem Fred set out to solve

Visual: what a drawing task looks like

the prompt

the model's drawing

what "correct" means

FRED'S DETAILS TO COME

Why drawing

Placeholder: why this task matters and where small models fall short.

What success means

Placeholder: how a drawing is judged, and the bar set in advance.

FRED'S EXPERIMENT

Fred's training journey

1 **Task**

to fill

2 **Data**

to fill

3 **Check**

to fill

4 **Train**

to fill

5 **Refine**

to fill

6 **Serve**

to fill

Decisions Fred made

teacher signal

student model

how drawings are checked

FRED'S DETAILS TO COME

What each step taught him

learning 1

learning 2

learning 3

FRED'S DETAILS TO COME

FRED'S EXPERIMENT

What Fred found, and how his approach compares

Results

before vs after
example drawings
cost to serve

FRED'S DETAILS TO COME

	SQL · FINANCE	MATHS	DRAWING · FRED
Checked by	**the grader**	**the grader**	**to fill**
Training signal	**teacher answers**	**own answers**	**to fill**
Student	**Qwen3-1.7B**	**Gemma 4 E2B**	**to fill**
Gain	**+21.6 / +28.4**	**+3.5**	**to fill**
Cheaper	**15.8× / 17.5×**	**4.8×**	**to fill**

WHAT BOTH APPROACHES SHOW

Placeholder: the lesson both approaches share.

SIMILAR APPROACHES

Meta's TorchTitan RL work found similar lessons

	TORCHTITAN RL · META	TPU-DISTIL
Task	long-horizon terminal agents; Terminal-Bench, 89 hard tasks	single-turn SQL, finance and maths answers
Student	Qwen3.5-9B	Qwen3-1.7B, Gemma 4 E2B
Hardware	8× B300: 2 trainer, 5 rollout, 1 eval workers	1 TPU v6e-8 host; teacher on 8× H200
Starting point	RL on ~1,000 cleaned tasks	distil verified answers, then RL
Algorithm	DPPO, loss averaged per prompt	GRPO, PPO clip, KL to the start
Sampling	asynchronous, never pauses; weights hot-swapped	synchronous rounds: sample, then update
Result	+52% on Terminal-Bench after 33 h of RL	+21.6 / +28.4 / +3.5 points; $79–$159

BOTH CLEANED THE DATA

Only 42% of Meta's tasks were clean before repair, and 40% of our teacher's BIRD answers were wrong.

BOTH CHECKED THE NUMERICS

A compiled kernel gave Meta a 2.5× log-prob mismatch. We check that our trainer and server agree.

BOTH REWARD FAILURES

Meta penalises a missing submission, and we score cut-off answers as wrong.

Meta figures from the speaker's notes of "Training long-horizon agents with TorchTitan RL", PyTorch Conference 2026; TPU-Distil figures from the README

TAKEAWAYS

What we learned

+22 / +28 / +3.5
points gained on SQL, finance and maths by students of about 2B parameters

4.8–17.5×
cheaper per correct answer than the 552B teacher

$79–$159
of compute to produce each student on one TPU host

If the student is weak, distil the teacher. The teacher's verified answers gave most of the gain.

Then add RL. Make sure the student's main failure gets a negative reward.

If the student is strong, let it learn from itself. Its own verified answers beat the teacher's.

Measure carefully. Compare models in one run, with a control and a noise floor, and price each correct answer.

How to run this recipe on your own task

github.com/jawadaminGOOG/tpu-distil

1 Measure

Score the teacher and a few candidate students with one prompt, and measure the noise floor.

`grade.py · gold + shuffled controls`

2 Generate

Sample the teacher 4 times and the student 8 times per question, at temperature 0.7.

`generate.py`

3 Verify

Keep the answers your grader accepts, up to 4 distinct per question, and audit for leaks.

`build_sft.py · leak_audit.py`

4 Distil

Train LoRA on both data sets (rank 256, lr 1e-4, batch 256, 3 epochs) and keep the better one.

`train_tpu_vm.sh sft`

5 Reinforce

Run GRPO with 8 samples at temperature 1.0, and count cut-off answers as wrong.

`build_grpo_rollouts.py --keep-truncated`

6 Serve and price

Serve on vLLM-TPU with one copy per chip, then work out the cost per correct answer.

`serve_small.sh · bench_serving.py`

How to choose the recipe. Train on the teacher's answers and on the student's own answers side by side. If the teacher's answers win, use KD + GRPO. If the student's own answers win, use self-distillation + GRPO, which needs no teacher time.

Code, settings and commands: github.com/jawadaminGOOG/tpu-distil (README recipe, docs/RECIPE.md)



Appendix

Backup slides for questions. Each one names the main slide it supports.

APPENDIX MAP

33 backup slides and the main slides they support

A1 The teacher ties up a whole server, even when id...

A2 How we price a correct answer → 4

A3 The prompt the student sees on BIRD → 5

A4 How the FinQA task is set up → 7

A5 How the MATH task is set up → 7

A6 A closer look at the two students → 7

A7 We fixed the evaluation design before seeing res...

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A22 All FinQA comparisons and the control → 15

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A25 How we score cut-off answers → 17

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A31 Adding your own task or student → 24

A32 A closer look at TorchTitan RL → 23

A33 What we'd borrow from TorchTitan RL → 23

A1 APPENDIX

Supports slide 4 · One frontier model needs a whole GPU server

The teacher ties up a whole server, even when idle

TEACHER DEPLOYMENT	
Endpoint	Vertex AI dedicated endpoint, a3-ultragpu-8g (8× H200), us-south1
Container	Model Garden vLLM `pytorch-vllm-serve`, release 20260922
Layout	tensor parallel 8 with expert parallelism, fp8 KV cache
Limits	65,536-token context, 128 concurrent sequences
Mode	non-thinking (`thinking: false`)
Cold start	about 25 minutes from zero; a keep-warm request every 5 minutes

38 min

of teacher generation the BIRD recipe needed (5.1 H200-hours)

5.3 h

that the endpoint stayed up during the BIRD work, costing 42.5 GPU-hours or $194

LESSON

A dedicated endpoint keeps billing while it sits idle. A batch job sized to the actual generation would cost far less.

docs/RECIPE.md (serving), docs/COMPUTE.md (endpoint uptime)

A2 APPENDIX

How we price a correct answer

Supports slide 4 · Every correct answer has a price

$ per 1,000 correct = 1,000 × $/hour ÷ (requests/s × 3,600 × accuracy)

requests/s: best error-free rate over a concurrency ladder, on the task's own prompts

MODEL, BIRD	$/HOUR	REQ/S	ACCURACY	$ / 1,000 CORRECT
Teacher, 8× H200	36.60	11.7	60.7%	1.438
Untrained, 1 v6e chip	2.70	13.8	30.6%	0.178
Trained student, 1 v6e chip	**2.70**	**15.8**	**52.2%**	**0.091**

The pricing favours the teacher

We price the TPU at its on-demand rate and the teacher at the cheapest listed H200 rate (DWS flex-start), without CPU or memory. At the same purchase tier the gap would roughly double.

How fast the teacher would need to be

The student loses its cost advantage only if the teacher served 15.8× faster on BIRD, 17.5× on FinQA or 4.8× on MATH.

Prices from the Cloud Billing Catalog, effective 2026-10-09

A3 APPENDIX

Supports slide 5 · Generic small models miss domain details

The prompt the student sees on BIRD

PROMPT TEMPLATE · SHA-256 RECORDED PER RUN

```
SYSTEM  You write SQLite queries.

USER    1  CREATE TABLE statements
        2  3 example rows per table,
           each cell cut to 40 chars
        3  BIRD external-knowledge hint
        4  the question
        5  reason briefly, then one
           ```sql``` block
```

BIRD DATA	
Evaluation	dev: 1,534 questions on 11 databases, never trained on
Training	64 of 69 training databases, 8,432 questions
Removed	`works_cycles`: 64,016-character schema
Hold-out	4 training databases, 522 questions, for screens
Grader	standard-library sqlite3, read-only copy, 30 s timeout

docs/RECIPE.md §1; system message paraphrased
```

A4 APPENDIX

How the FinQA task is set up

Supports slide 7 · Each task has an automatic grader

PROMPT FINQA-2 · SHA-256 RECORDED

```
SYSTEM  You are an expert financial
        analyst.

USER    report text before the table
        the table, pipe-separated rows
        report text after the table
        the question
        reason briefly, end with
        Answer: <number>
        (yes/no only for yes/no questions)
```

FINQA	
Splits	train 6,251 · dev 883 · test 1,147 (evaluation)
Leak filter	142 training questions removed (exact or 13-word overlap); no shared report pages
Grader	within 1%, or 0.005 absolute below 0.5; ×100 and ÷100 forms
Controls	gold 100% · shuffled 0.6%

LESSON

Our first prompt offered a yes/no answer on every question, and the base student answered yes or no on 363 numeric ones. We replaced the prompt, using dev, before generating any teacher answers.

docs/RECIPE.md §8; README

A5 APPENDIX

How the MATH task is set up

Supports slide 7 · Each task has an automatic grader

PROMPT MATH-1 · SHA-256 RECORDED

```
SYSTEM   You are an expert
         mathematician.

USER     the problem
         reason briefly, end with
         \boxed{<answer>}

GRADE    last \boxed{...}, balanced braces
         reference strip_string normalisation:
         spacing, \left/\right, dfrac, units,
         degrees, percent, 0.5 = \frac{1}{2}
```

MATH

Source	EleutherAI/hendrycks_math, MIT licence; 7 subjects, 5 levels
Splits	test 5,000 (evaluation) · dev 500 held out by seed, by subject
Leak filter	13 identical and 725 13-word overlaps removed: 6,262 remain
Controls	gold 100% · shuffled within subject 0.0%

CAVEAT

Exact matching counts equal answers written differently, like 2√2 and √8, as wrong. This affects every model equally.

docs/RECIPE.md §9; README

A6 APPENDIX

A closer look at the two students

Supports slide 7 · One student starts weak, the other starts strong

	QWEN3-1.7B	GEMMA 4 E2B-IT
Used on	BIRD, FinQA	MATH
Size	1.41B decoder matmul + 0.31B tied head	35 layers, about 2B effective
LoRA	rank 256 on 7 projections, 196 modules, 0.28B trainable	rank 256, 190 modules incl. fused k/v
Base accuracy	30.6% BIRD · 44.0% FinQA	74.9% MATH
Teacher	60.7% BIRD · 75.7% FinQA	92.5% MATH
Licence	open weights	Apache-2.0

Qwen3-4B

It already scores 61.7% on BIRD without training. We didn't measure its serving cost.

LESSON

Picking the right student can matter as much as distilling it.

README, docs/COMPUTE.md, docs/RECIPE.md §9

A7 APPENDIX

Supports slide 8 · Six decisions shaped the experiments

We fixed the evaluation design before seeing results

Targets set in advance

We wrote down the pass conditions before each stage.

Test sets stayed untouched

No evaluation set chose a checkpoint or a hyperparameter, and we fixed the final checkpoint in advance.

One run per comparison

We serve and grade every compared model together, with greedy decoding and a 2,048-token limit.

Two copies of the baseline

Every run includes two copies of the untrained model, which give the noise floor.

Paired statistics

We compare models question by question and report bootstrap 95% CIs from 10,000 resamples.

Early stopping

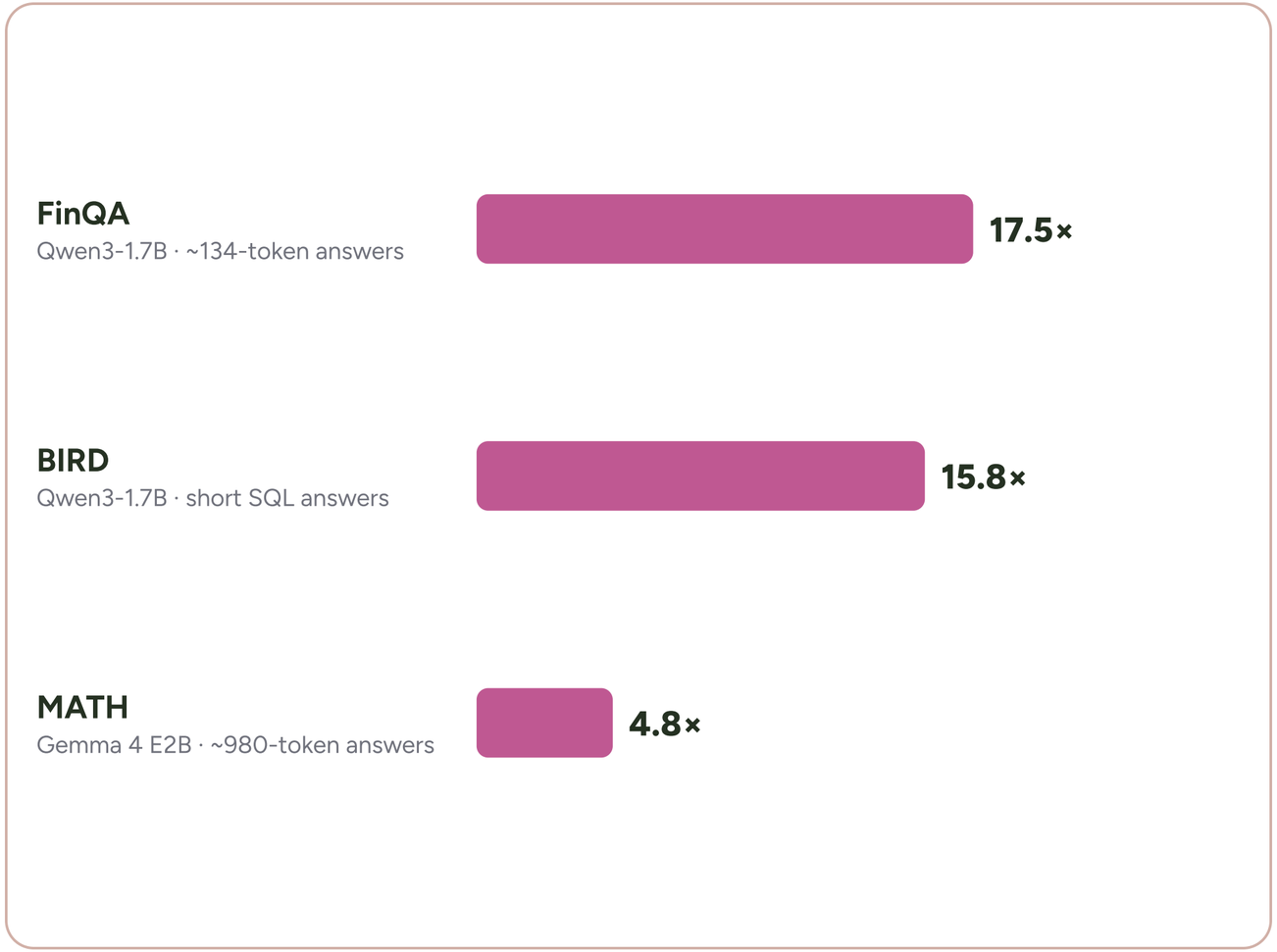
BIRD used only its 522 held-out training-database questions for early stopping.

docs/RECIPE.md §7, README limits

A8 APPENDIX

Answer length is a serving cost

Supports slide 8 · Six decisions shaped the experiments



Bars: teacher ÷ student cost per correct answer (README)

Why we turned thinking off

Teacher and student both answer briefly. The student would copy long reasoning traces, and the teacher samples 4× faster without them.

LESSON

On MATH the student's answers are 2.3× longer than the teacher's, so one chip finishes fewer requests than eight GPUs, and throughput drops to 0.42×.

A9 APPENDIX

Every grader passed two checks

Supports slide 11 · Check the measurements before the experiment

TASK	GOLD ANSWERS	SHUFFLED ANSWERS	HOW AN ANSWER IS GRADED
BIRD	100% of 1,530 executable	0.2%	run the last SQL block on a read-only copy; compare row sets
FinQA	100%	0.6%	number after the last `Answer:`, within 1%; ×100 and ÷100 forms accepted
MATH	100%	0.0%	last `\boxed{}` after the reference normalisation

LESSON

Before any accuracy number means anything, the grader has to score gold answers at 100% and shuffled answers near 0%. FinQA's grader is lenient, so its scores aren't comparable to published FinQA results.

README, docs/RECIPE.md

A10 APPENDIX

Supports slide 11 · Choose who writes the training answers

How much training data each task produced

SET	QUESTIONS	SAMPLES	CORRECT	ROWS KEPT	TIME
BIRD, teacher ×4	8,432	33,728	20,133 (60%)	10,112	38 min
FinQA, teacher ×4	6,109	24,436	18,487 (76%)	6,016	26 min
MATH, teacher ×4	6,262	25,048	93.5%	4,992 of 6,021	—
MATH, student ×8	6,262	50,096	72.6%	4,992 of 5,237	no teacher

How we sampled

We sampled at temperature 0.7 (top-p 0.95 on FinQA and MATH), in non-thinking mode, on training questions only. Each control uses the same number of the student's own answers.

NOTE

On MATH, all correct answers to a problem reduce to the same normalised answer, so the 4-distinct limit keeps one row per problem.

docs/COMPUTE.md, docs/RECIPE.md §4, §8, §9

A11 APPENDIX

The leak audit found nothing

Supports slide 12 · Keep only the answers the grader accepts

THE AUDIT REJECTS

Evaluation groups	BIRD databases, FinQA report pages, MATH problem fingerprints
Question overlap	exact match or a shared 13-word sequence with an evaluation question
Non-training ids	any id outside the training split
Stale answers	any kept answer that no longer verifies on re-grading
In the trainer	ids from `--eval_tasks` are refused again

0 leaks

found on BIRD, FinQA and MATH, across rows, groups, overlaps and re-grades

Some questions were removed up front

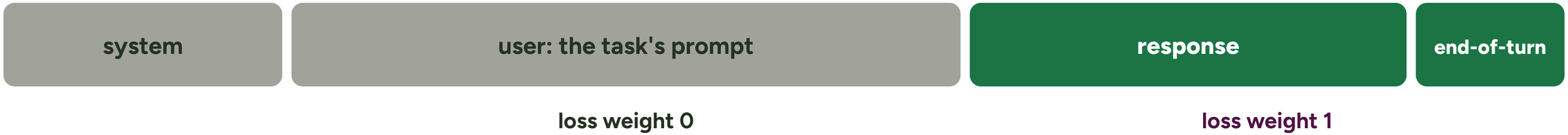
We removed 142 FinQA training questions and 738 MATH problems before generation. Much of MATH's wording is shared boilerplate, so its filter is strict.

docs/ARCHITECTURE.md, docs/RECIPE.md

A12 APPENDIX

Supports slide 12 · Keep only the answers the grader accepts

From graded answers to training records



ONE JSONL RECORD

```
{"id": "...#s0",
 "db_id" | "report" | "problem_hash",
 "messages": [system, user],
 "response": "..."}

GRPO adds: "group_id", "reward",
           "truncated" (cut-off answers)
```

BUILDER RULES

Distinct	at most 4 distinct answers per question
Exact N	drawn to a fixed size with a fixed seed
Template check	records the chat template alters are dropped and counted
Cut-off answers	no end-of-turn token in the loss

docs/ARCHITECTURE.md (record format)

A13 APPENDIX

Supervised fine-tuning settings

Supports slide 12 · Distil the student on one TPU host

SETTING	VALUE
Student weights	bf16, frozen
LoRA	rank 256, alpha 512, attention and MLP projections, every layer; float32 adapters
Batch	256 sequences: micro-batch 8 × gradient accumulation 32
Optimiser	AdamW, lr 1e-4, 5% warm-up, cosine to 0, clip 1.0
Length	≤ 8,192 tokens; buckets 2,048 / 4,096 / 8,192
Schedule	3 epochs: 117 steps (BIRD), 69 (FinQA), 57 (MATH)
Mesh	FSDP 8 × TP 1; rematerialisation; splash attention

10–27 min

SFT wall time on the v6e-8: MATH 10, FinQA 13, BIRD 27

RULE

Every size is a multiple of 256 to match the v6e's 256 × 256 matrix unit.

docs/RECIPE.md §5, docs/COMPUTE.md

A14 APPENDIX

Supports slide 12 · Distil the student on one TPU host

The trainer and the server compute the same model

0.998

Qwen3 top-1 agreement, trainer vs vLLM, 16 prompts (limit 0.99); KL 1e-4

0.994

Gemma 4 top-1 agreement on vLLM's own continuations; KL 5.8e-4

1 : 1

retrain: identical loss at all 117 steps, byte-identical checkpoint

The prompts match token for token

A test on the real tokenizer checks that the training context equals the prompt vLLM serves; Gemma also passes vLLM's /tokenize check.

Every model uses the same flags

Merged checkpoints serve with the base model's flags, so compared models differ only in weights.

trainer/parity_check.py; docs/RECIPE.md §5, §9

A15 APPENDIX

What it took to add Gemma 4

Supports slide 12 · Distil the student on one TPU host

PROBLEM	FIX
vLLM-TPU warm-up fails on the default PyTorch model path	`MODEL_IMPL_TYPE=flax_nnx`: the JAX Gemma 4 model
System message rendered as a Python list in 8 of 16 prompts	`--chat-template-content-format string`; earlier outputs rerun
vLLM refuses prompt log-probs for multimodal models	parity on vLLM's own 64-token continuation
qwix traces LoRA on a 2-token input splash attention refuses	trace with flash attention off, then restore it

Trainer changes

Tunix's Gemma 4 model handles per-layer embeddings and KV sharing. LoRA covers the fused k/v in the 15 layers that compute K/V, and the logit soft-cap runs in float32.

LESSON

Adding a new model family took one loader, one exporter and four serving fixes. The first start compiles for about 17 minutes per model.

docs/RECIPE.md §9, problems and fixes

A16 APPENDIX

What happens in one GRPO round

Supports slide 13 · RL makes sometimes-right answers always right

1 Serve

7–8 replicas of the current student

2 Sample

8 answers per question
T 1.0, top-p 1.0

3 Grade

reward 1 if verified, else 0

4 Select

≤ 2,048 questions with mixed rewards

5 Update

64 sequences per step one pass

SETTING	VALUE	SETTING	VALUE
Advantage	group-relative ÷ group std	Optimiser	AdamW; lr 2e-6 rounds 1–2, 5e-6 rounds 3–6
Policy loss	PPO clip 0.2, dual clip 3	Length	≤ 8,192 tokens
KL	k3 vs the starting checkpoint, β 0.02	Guard	stop if a step's KL > 0.2

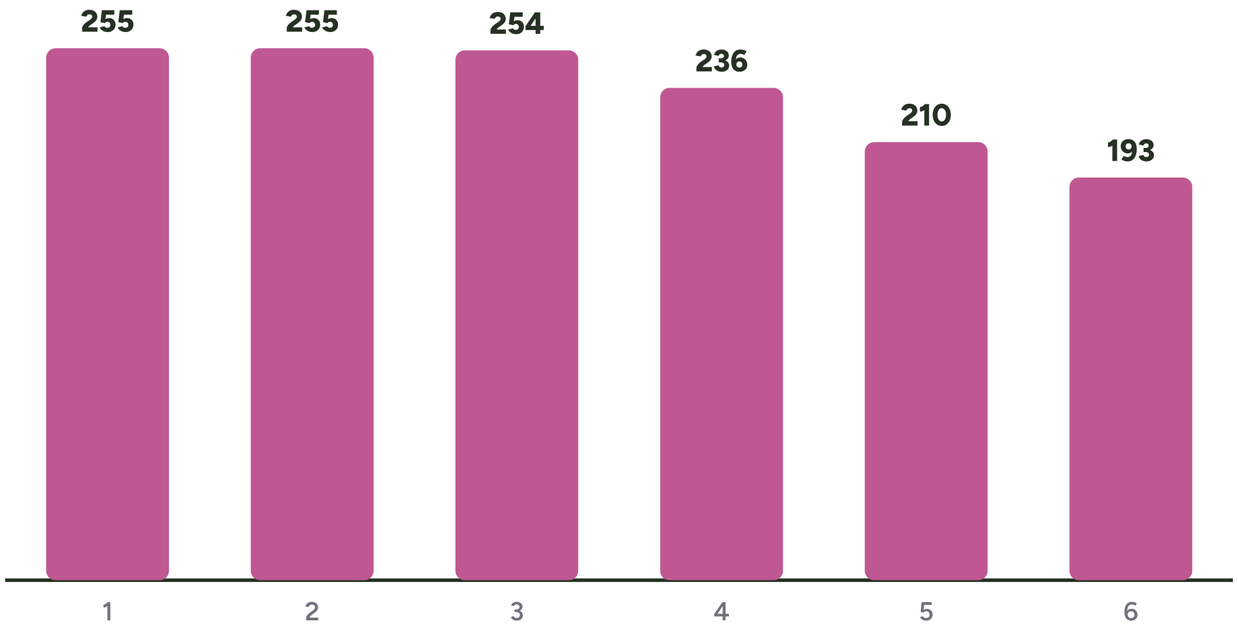
docs/RECIPE.md §6

A17 APPENDIX

RL stayed stable

Supports slide 13 · RL makes sometimes-right answers always right

FinQA steps per round fewer questions have mixed outcomes each round



≤ 0.05

the highest KL in any step, round or task (BIRD 0.033, FinQA 0.049, MATH 0.017)

0

times the KL guard (0.2) had to stop a run

LESSON

At lr 2e-6, two rounds gave 1.7 to 2.1 points on BIRD. Raising it to 5e-6 for four more rounds reached 4.0.

docs/RECIPE.md §6, §8, §9

A18 APPENDIX

What RL doesn't do

Supports slide 13 · RL makes sometimes-right answers always right

+0.4

BIRD RL gain on 522 held-out training-database questions, CI [−2.3, +2.9], which is too small to detect

73.1 → 74.3

BIRD best-of-8 on the training pool across 5 rounds, almost unchanged

133 → 159

mean FinQA answer tokens under RL (+19%)

LESSON

RL sharpened answers the student could already reach instead of teaching new ones. GRPO added 1.82 points to the MATH KD student but only brought it back level with base, so RL can't make up for the wrong SFT data.

README; distil/passk.py

A19 APPENDIX

How we measured serving throughput

Supports slide 13 · Each student runs on a single chip

	STUDENT	TEACHER
Software	vLLM 0.26.1rc1 TPU build	Model Garden vLLM
Layout	1 replica per v6e chip, TP 1	TP 8 + expert parallel
KV cache	bf16	fp8
BIRD best	15.8 req/s	11.7 req/s, flat 128 → 512
FinQA best	24.1 req/s, still rising at 256	17.9 req/s at 128
MATH best	2.79 req/s at 128	6.64 req/s at 128

Method

We raised concurrency until errors appeared and priced the best error-free rate. Each model ran in its own session on the same hardware and build.

NOTE

The BIRD and FinQA benchmarks cap answers at 1,024 tokens. MATH uses 2,048, because a 1,024 cap would cut off 36% of the student's answers.

README serving tables, docs/RECIPE.md §2

A20 APPENDIX

What broke, and how we fixed it

Supports slide 13 · Each student runs on a single chip

PROBLEM	FIX
Out of memory with two 8,192-token sequences per chip	one sequence per chip; batch of 256 kept
A prompt log-prob request (≈ 4.6 GB extra) stopped a serving replica	parity runs on a separate replica with less cache
A chip's runtime stopped: SLICE_FAILURE_WORKER_UNAVAILABLE	a watchdog restarts replicas; the client retries
Two copies of the same weights gave different text on 45% of FinQA questions	same-run comparisons; two base copies per run
Teacher takes about 25 minutes to start from zero	keep-warm loop during teacher stages
The TPU VM was unavailable for 7 hours during MATH rollouts	disclosed; a driver should stop when any rollout fails

docs/RECIPE.md, problems and fixes

A21 APPENDIX

Supports slide 15 · All three students improved

All BIRD comparisons and training-data options

BIRD DEV, SAME RUN	Δ POINTS [95% CI]	BETTER / WORSE
KD + GRPO − base	**+21.6** [+19.2, +24.1]	387 / 55
KD − base	**+17.6** [+15.2, +20.1]	334 / 64
KD + GRPO − KD	**+4.0** [+2.2, +5.9]	136 / 74
Base − base copy	+0.3 [−0.8, +1.4]	39 / 34
Teacher − KD + GRPO	+8.5 [+6.0, +10.9]	261 / 131

TRAINING DATA (SEPARATE RUN)	BIRD
KD: 10,112 teacher answers	**48.7%**
CORR: 5,883 teacher corrections	37.1%
Self: 10,112 own answers	35.6%
Base	30.6%

README

A22 APPENDIX

All FinQA comparisons and the control

Supports slide 15 · All three students improved

FINQA TEST, SAME RUN	Δ POINTS [95% CI]	BETTER / WORSE
KD + GRPO − base	**+28.4** [+25.5, +31.5]	367 / 41
KD − base	**+22.6** [+19.4, +25.7]	321 / 62
KD + GRPO − KD	**+5.8** [+3.9, +7.8]	100 / 33
Base − base copy	−1.9 [−3.3, −0.5]	23 / 45
Teacher − KD + GRPO	+3.2 [+1.0, +5.4]	105 / 68

The control (earlier run)

Self-distillation scored 57.0% and KD 66.6%. KD beat self by 9.6 points [+6.9, +12.4], and self beat base by 14.6 [+11.8, +17.4].

WHY THE CONTROL GAINS MORE

The teacher's share is smaller than on BIRD because part of FinQA's self-distillation gain comes from learning the answer format. The base student gives no number at all on more than 100 test questions.

README

A23 APPENDIX

Every MATH result

Supports slide 16 · A strong student lost accuracy copying the teacher

SAME-RUN EVALUATION, MATH TEST	ACCURACY	Δ [95% CI]	CUT-OFF ANSWERS
Base, two copies	74.90 / 74.78	−0.12 [−0.88, +0.62]	—
KD: 4,992 teacher answers	72.56	−2.34 [−3.44, −1.22] vs base	577 (base 800)
Self-distillation: 4,992 own answers	76.84	+1.94 [+1.08, +2.84] vs base	—
Self + GRPO, cut-off scored wrong	78.38	+1.82 [+1.00, +2.66] vs self	674
Self + GRPO, cut-off dropped	77.40	+0.84 [+0.02, +1.68] vs self	756
KD + GRPO	≈ base	+1.82 [+0.80, +2.86] vs KD	—
Self + GRPO − KD + GRPO (separate run)		+2.82 [+1.78, +3.88]	

README, MATH section

A24 APPENDIX

Teacher corrections hurt on MATH

Supports slide 16 · A strong student lost accuracy copying the teacher

CORR ON MATH	
Input to the teacher	the student's first failed answer and one sentence about it; never the gold answer
Failures	2,793 problems: 1,620 cut off, 653 wrong answer, 520 no boxed answer
Rewrites	4 per problem, 74.9% correct, 2,362 rows
Result	**64.26%**: −10.76 [−11.96, −9.54] vs base
Length	559 tokens on average vs self-distillation's 1,013

8%

of corrected answers carry correction language: "the student's error was…"

LESSON

The corrections are short, written in the teacher's style, and come only from hard problems. They pull a strong student furthest from its own reasoning.

README; docs/RECIPE.md §9

A25 APPENDIX

How we score cut-off answers

Supports slide 17 · Counting cut-off answers as wrong made RL work

BUILD_GRPO_ROLLOUTS.PY --KEEP-TRUNCATED

```
rollout hits 2,048 tokens
  no end-of-turn token emitted

before: dropped from its group
  -> the main failure never gets
     a negative reward

now:    kept, reward 0,
        "truncated": true
  -> data.py gives the trailing
     end-of-turn token weight 0
```

~600 → ~1,300

MATH problems per round with mixed outcomes

+0.98

scored wrong vs dropped, CI [+0.16, +1.78]

WHY WE DROP THE END-OF-TURN TOKEN

If we trained on a cut-off answer as written, the student would learn to stop where it never stopped. Leaving out the end-of-turn token lowers the probability of exactly the tokens it wrote.

docs/RECIPE.md §9 (truncated rollouts)

A26 APPENDIX

Supports slide 18 · The trained student is both cheap and accurate

Cost per correct answer on each task

TASK	HARDWARE PRICE	× THROUGHPUT	× ACCURACY	= CHEAPER	STUDENT VS TEACHER $ / 1,000 CORRECT
BIRD	13.6×	1.35× (15.8 vs 11.7 req/s)	0.86×	**15.8×**	0.091 vs 1.438
FinQA	13.6×	1.35× (24.1 vs 17.9 req/s)	0.96×	**17.5×**	0.043 vs 0.750
MATH	13.6×	0.42× (2.79 vs 6.64 req/s)	0.85×	**4.8×**	0.343 vs 1.656

LESSON

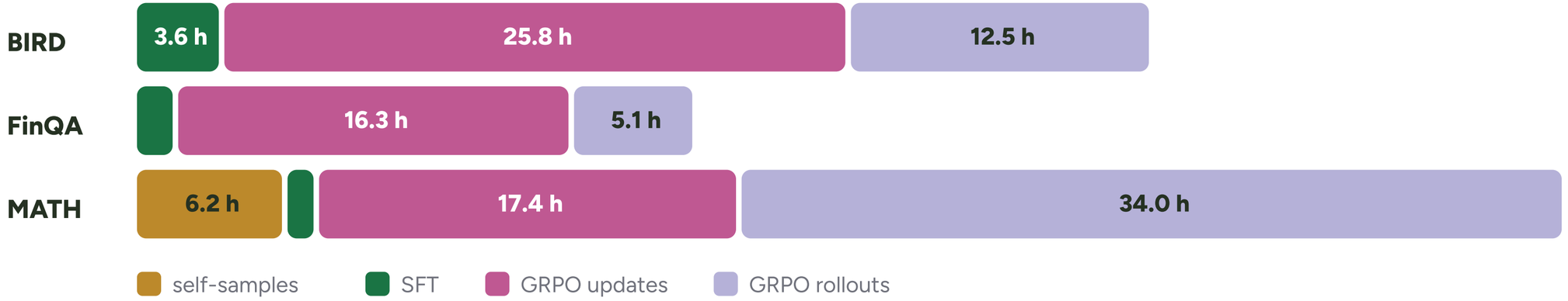
Hardware price gives the same 13.6× on every task, and throughput decides the rest. Short answers keep throughput above 1×, while long MATH answers cut it to 0.42×.

README serving tables

A27 APPENDIX

Supports slide 18 · The trained student is both cheap and accurate

How much compute each task used



$136

BIRD: 41.9 chip-hours + 5.1 teacher H200-hours

$79

FinQA: 23.1 chip-hours + 3.5 teacher H200-hours

$159

MATH: 58.9 chip-hours, no teacher time

docs/COMPUTE.md, generated by distil/compute_report.py; bars in TPU v6e chip-hours

A28 APPENDIX

What these results don't show

Supports slide 18 · Why we trust these results

One training seed

We trained each model once, so the noise floor measures serving noise, not training-seed variance.

Student and task changed together

The student changed along with the task, so "already strong" is our best explanation, not a proven cause.

Public benchmarks

The teacher and students may have seen BIRD, FinQA or MATH during pre-training. This applies to every model.

The FinQA grader is lenient

It accepts a 1% tolerance and percentage forms, so our scores aren't comparable to published FinQA results.

We missed the MATH targets

We required KD to gain 5 points (it lost 2.3) and each GRPO stage to gain 2 (the best gained 1.82). A 7-hour outage cut one round to 3,367 of 6,262 problems.

The cost gap depends on deployment

A faster teacher deployment would shrink the cost advantage (see A2).

README, what this does not show

A29 APPENDIX

Versions and machines

Supports slide 10 · We built everything on open-source libraries

LIBRARY	VERSION	USE
vLLM (TPU build)	0.26.1rc1	student serving
JAX · libtpu	0.11.0 · 0.0.44	compile, TPU runtime
google-tunix	0.1.8	Qwen3 / Gemma 4 models, trainer
qwix	0.1.8	LoRA adapters
Flax NNX · Optax · Orbax	0.12.8 · 0.2.6 · 0.12.2	modules, AdamW, checkpoints
transformers · safetensors	5.12.1 · 0.7.0	tokenizer, template, files

MACHINE	ROLE
TPU v6e-8 VM, us-east5-a	serves, trains, merges, benchmarks the student; never serving and training at once
CPU VM n4-standard-64	databases, prompts, grading, data builds, long-run drivers
Vertex AI endpoint, 8× H200, us-south1	the teacher
Cloud Storage bucket	checkpoints, data, evidence

docs/RECIPE.md, docs/ARCHITECTURE.md

A30 APPENDIX

How the code is organised

Supports slide 10 · We built everything on open-source libraries

1 prompts.py
versioned template per task

2 generate.py
teacher or student resumes by id

3 grade.py
task grader gold + shuffled

4 build_*.py
SFT or GRPO records + leak_audit.py

5 trainer/
SFT / GRPO / merge Tunix on v6e-8

6 report.py
paired CIs, cost from raw files

distil/

Standard-library scripts. Every task-specific step goes through `distil/tasks/` (bird, finqa, math), selected with `--task`.

trainer/

`train.py` picks Qwen3 or Gemma 4 from config.json, `data.py` masks the loss, and `parity_check.py` and `kl_guard.sh` check the results.

docs/ARCHITECTURE.md

A31 APPENDIX

Adding your own task or student

Supports slide 24 · How to run this recipe on your own task

A NEW TASK IS ONE MODULE

```
# distil/tasks/<your_task>.py
load_items(split)      -> items
PROMPT_VERSION, render(item)
extract(response)      -> answer
grade(answer, item)    -> bool

# controls before use
gold answers   ~ 100%
shuffled       ~ 0%

# then: add the name to NAMES
```

Adding a new student

Add a config builder, loader, LoRA targets and delta exporter in `student_models.py`, and its end-of-turn marker in `data.py`.

BEFORE YOU TRAIN

Check logit parity with parity_check.py, and check the served chat template with vLLM's /tokenize. We added FinQA, MATH and Gemma 4 this way.

docs/ARCHITECTURE.md, extension points; interface names abbreviated

A32 APPENDIX

A closer look at TorchTitan RL

Supports slide 23 · Meta's TorchTitan RL work found similar lessons

Sampling never pauses

Rollouts run continuously, new weights are swapped in while they run, and the buffer takes any 32 finished groups in any order.

The KV cache survives updates

They keep it across weight updates with no accuracy loss. KV headroom limits how many rollouts run in parallel.

The agent runs outside the sandbox

The policy controls a sandbox but doesn't run inside it.

Each failure gets its own reward

A dead sandbox gets no penalty, a missing submission gets a penalty, and hitting the context limit gets a nudge.

They cleaned the tasks

Only 42% of TMAX tasks and 61% of CalibForge tasks were clean before repair.

DPPO with per-prompt averaging

They use DPPO instead of GRPO or PPO, and average the loss per prompt instead of per token.

Speaker's notes of "Training long-horizon agents with TorchTitan RL", PyTorch Conference 2026

A33 APPENDIX

Supports slide 23 · Meta's TorchTitan RL work found similar lessons

What we'd borrow from TorchTitan RL

IDEA FROM TORCHTITAN RL	WHERE IT FITS IN TPU-DISTIL	WHY
Asynchronous rollouts	overlap sampling and updates across the 8 chips	rollouts were 30% of BIRD and 58% of MATH chip-hours, with the trainer idle
Online task generation (+8 points vs a static pool)	generate new questions per round from training data	later FinQA rounds ran out of mixed-outcome questions
Failure-type rewards	extend the MATH cut-off rule to other failure types	each failure mode needs its own signal
Bitwise trainer–sampler parity	tighten parity from 0.998 top-1 toward exact	removes a source of off-policy drift
Per-prompt loss averaging	an option in the GRPO loss	long answers stop dominating the update

These are ideas; we haven't tested any of them in TPU-Distil